

## SQL Learning Journal #6 – Aggregate Functions to

### Understand Data

#### Introduction

In today's lesson, I learned how to use SQL aggregate functions to summarize data. I practiced

AVG(), SUM(), MIN(), MAX(), COUNT(), and ROUND() functions and learned how to combine them

with filtering conditions.

#### 1. Aggregate Functions

AVG(), SUM(), MIN(), MAX(), and COUNT() are aggregate functions used after SELECT. AVG() and SUM() work only with numerical values.

#### 2. MIN(), MAX(), and COUNT()

MIN(), MAX(), and COUNT() work with numerical and text values. MIN() returns the first text alphabetically; MAX() returns the last.

#### 3. Using Aliases

```
SELECT MAX(field_name) AS nick_name_max FROM table_name;
```

#### 4. Combining Filtering with Aggregate Functions

```
SELECT AVG(year) AS average_year FROM table_name WHERE city='Istanbul';
```

```
SELECT SUM(year) AS sum_year FROM table_name WHERE city='Istanbul';
```

#### 5. Using ROUND()

ROUND() rounds decimal values. Omitting the second argument is the same as using 0.

#### 6. ROUND() with Positive and Negative Values

Positive values round digits to the right of the decimal point. Negative values round digits to the left.

#### What I Learned Today

I learned how to summarize data using aggregate functions and combine them with filtering conditions. Careful use of commas and parentheses is essential.